

0 2

A student investigated the distribution of stomata on leaves from two species of plant. She removed small pieces from the lower surface of the leaves of each plant species. She mounted these pieces on separate microscope slides. She then counted the number of stomata in several parts of the epidermis on each piece of leaf tissue using an optical microscope.

0 2 . 1

Suggest appropriate units the student should use to compare the distribution of stomata on leaves.

[1 mark]

0 2 . 2

The pieces of leaf tissue examined were very thin.

Explain why this was important.

[2 marks]

0 2 . 3

Give **two** reasons why it was important that the student counted the number of stomata in several parts of each piece of leaf tissue.

[2 marks]

1

2



0 2 . 4

One of the two plant species used by the student in this investigation was a xerophyte.

Other than the distribution of stomata, suggest and explain **two** xerophytic features the leaves of this plant might have.

[2 marks]

1 _____

2 _____

0 2 . 5

The student then compared the rate of transpiration (evaporation of water) from the two species of plant. She did this by measuring the rate of water uptake by each plant species.

Suggest **two** reasons why the rate of water uptake by a plant might **not** be the same as the rate of transpiration.

[2 marks]

1 _____

2 _____

Turn over for the next question

